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Field-Scale Storage of LOHCs in Depleted Reservoirs: Energy Storage and Enhanced Oil Recovery

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Abstract

Liquid Organic Hydrogen Carriers (LOHC), such as Toluene/Methylcyclohexane (MCH), offer a reversible, safe, and efficient method for storing and transporting hydrogen. This study investigates the dual potential of LOHC injection in depleted hydrocarbon reservoirs—enabling cyclic hydrogen storage while unlocking incremental oil recovery. By integrating energy storage with enhanced oil recovery (EOR), this approach aims to improve the feasibility of subsurface hydrogen initiatives, particularly in regions with existing petroleum infrastructure. High-resolution compositional reservoir simulations were performed using the commercial compositional simulator to evaluate the field-scale performance of LOHC storage in a representative depleted reservoir. Toluene was injected and later extracted as part of a reversible hydrogen cycle, accounting for its conversion to/from MCH. Multiple cyclic injection–withdrawal scenarios were modeled to mimic real-world operational conditions and evaluate key metrics, including LOHC recovery, pressure response, mass transfer, and oil mobilization. Thermodynamic behavior, fluid interactions, and flow properties were calibrated using available PVT data. A base model was constructed with realistic petrophysical properties (porosity, permeability, and initial saturation), and injection-production strategies were optimized to enhance recovery efficiency while maintaining reservoir stability. The reservoir response to each operational cycle was assessed in terms of fluid displacement efficiency, water cut, and compositional tracking of the LOHC components. Simulation results indicate that optimized cyclic injection strategies led to a LOHC recovery factor of approximately 75%, demonstrating the technical feasibility of reversible hydrogen storage in depleted formations. Moreover, injecting Toluene led to favorable thermodynamic and displacement conditions that enabled up to 40% incremental oil recovery from the depleted zones. The cyclic operation maintained effective pressure support across simulation periods, while water cut was consistently less than 0.0001%, confirming highly selective displacement behavior. These results suggest that LOHC injection not only supports the hydrogen economy through secure subsurface storage but also rejuvenates mature fields with additional oil production.

Keywords: Liquid Organic Hydrogen Carriers (LOHC), Underground Hydrogen Storage (UHS), Brown fields, Enhanced Oil Recovery (EOR), Unlocking incremental Oil

Introduction

The global energy landscape is undergoing a transformative shift toward sustainable and low-carbon solutions, driven by the urgent need to mitigate climate change and reduce dependence on fossil fuels (Alanazi et al., 2025; Han et al., 2024; Tariq et al., 2025). This transition emphasizes cleaner energy production, efficient storage technologies, and the integration of renewable sources such as solar and wind into the global energy mix (Feng et al., 2025; Janjua et al., 2025, 2024; Tariq et al., 2023a). At the same time, achieving sustainability requires scalable systems that can ensure both energy security and environmental stewardship, particularly in the context of growing global energy demand (Ferlin et al., 2021; He et al., 2015; Sage et al., 2024). Within this framework, hydrogen has emerged as a promising energy carrier due to its high energy density, versatility, and potential to decarbonize sectors such as transportation, industry, and power generation (Tariq et al., 2023b). However, the storage and transportation of hydrogen present significant technical and economic challenges, primarily due to its low volumetric energy density and the safety concerns associated with its handling in gaseous or cryogenic liquid forms. Traditional methods, such as compressed hydrogen (CH_2) and liquefied hydrogen (LH_2), face significant operational challenges due to extreme pressure and temperature requirements (Clematis et al., 2023). Consequently, alternative hydrogen carriers, such as Liquid Organic Hydrogen Carriers (LOHCs) have gained increasing attention as a viable solution to these challenges, offering a safe, efficient, and scalable method for hydrogen storage and transport (Chu et al., 2023; D'Ambra and Gébel, 2023).

LOHCs are organic compounds capable of reversibly storing and releasing hydrogen through catalytic hydrogenation and dehydrogenation processes (Kim et al., 2022). During hydrogenation, hydrogen is chemically bonded to the carrier molecule for storage; it is subsequently released via dehydrogenation, with the carrier remaining chemically stable over multiple cycles (Rüde et al., 2022). Unlike conventional hydrogen storage methods—such as compressed gas or liquefied hydrogen—LOHCs allow hydrogen to be stored in a liquid state under ambient conditions, enabling the use of existing liquid fuel infrastructure for transportation and distribution (Niermann et al., 2019; Rao and Yoon, 2020). Among the various LOHC systems, the Toluene/Methylcyclohexane (MCH) pair is particularly promising due to its high chemical stability and hydrogen storage capacity of approximately 6.1 wt.% (Bárkányi et al., 2024; Cho et al., 2021). In this system, hydrogen is stored in the hydrogen-rich molecule (MCH) and later released through catalytic dehydrogenation, regenerating the original compound (Toluene) for repeated use (Spatolisano et al., 2024). These features make LOHCs especially attractive for long-distance hydrogen transport, seasonal storage, and large-scale integration into existing energy supply chains.

The appeal of LOHCs lies in their compatibility with existing liquid fuel infrastructure, which significantly reduces the capital investment required for large-scale deployment. Pipelines, tankers, and storage facilities designed for petroleum products can be repurposed for LOHC transport, offering a practical pathway for integrating hydrogen into global energy markets. Furthermore, LOHCs exhibit favorable safety characteristics, as they are non-explosive, non-toxic, and have low vapor pressures, minimizing the risks associated with hydrogen handling (Bárkányi et al., 2024; Reuß et al., 2017). These attributes make LOHCs particularly attractive for long-distance hydrogen transport and seasonal energy storage, addressing key bottlenecks in the hydrogen supply chain. Simultaneously, the energy sector continues to seek economically viable methods for enhancing oil recovery from mature and depleted fields. Enhanced oil recovery (EOR) through gas or chemical injection has proven effective but often comes with significant operational and environmental costs. In this context, the integration of LOHC injection into subsurface operations offers a dual opportunity: storing hydrogen in a chemically stable form while mobilizing residual hydrocarbons via miscible displacement and re-pressurization. Such a hybrid LOHC–EOR strategy could unlock substantial residual oil volumes while enabling large-scale, reversible hydrogen storage using existing geological infrastructure.

Despite the conceptual promise, the practical implementation of LOHC-based hydrogen storage in depleted oil fields remains largely unexplored. Key knowledge gaps include the thermodynamic and phase behavior of LOHCs under reservoir conditions, compositional transport mechanisms, pressure and injectivity response, and long-term recovery efficiency. Therefore, this study addresses this knowledge gap by conducting a comprehensive compositional reservoir simulation using CMG-GEM to assess the field-scale performance of Toluene-based LOHC storage in a heterogeneous, depleted sandstone reservoir. The work evaluates multiple cyclic injection–withdrawal scenarios, examining LOHC recovery, pressure behavior, fluid saturation dynamics, and the extent of incremental oil recovery.

Methodology

Numerical Simulation Study

A commercial compositional reservoir simulator (CMG-GEM) was used to evaluate the feasibility of storing LOHC and simultaneously unlocking incremental oil in a depleted sandstone reservoir. The model was constructed using a 3D corner-point grid system comprising 138,249 cells, discretized into $43 \times 51 \times 63$ blocks in the x, y, and z directions, respectively. The model spans a physical domain of 2500 meters in length (x-direction), 3000 meters in width (y-direction), and 100 feet in vertical (z-direction) to simulate a representative sector of the actual reservoir. Initial reservoir conditions included about to deplete average pressure of 2,500 psi, a temperature of 180°F, and reservoir salinity of 57,100 ppm. The reservoir rock compressibility was defined as $3.0 \times 10^{-6} \text{ psi}^{-1}$, and an average porosity of 24.5% was assigned. The water phase had a salinity of 57,100 ppm, and the irreducible water saturation (S_{wi}) was set at 0.19, ensuring realistic initial conditions for multiphase flow. The simulation scenario involved cyclic injection and withdrawal of Toluene, modeled as a representative LOHC. Each operational cycle spanned one year, divided into three phases:

- Injection phase: 5 months of continuous LOHC injection at a rate of 2,000 bbl/day
- Soaking phase: 2 months of shut-in period for pressure redistribution
- Withdrawal phase: 5 months of fluid production at 2,000 bbl/day

Governing Equations

The compositional reservoir simulations were governed by the multiphase, multi-component flow equations implemented in CMG-GEM, which solve for mass conservation of each component across the reservoir domain. The generalized form of the governing equation is given by Eq. (1),

$$\frac{\partial}{\partial t} \left(\phi \sum_{\alpha} S_{\alpha} \rho_{\alpha} x_{\alpha,i} \right) - \nabla \cdot \left\{ K \sum_{\alpha} \frac{k_{r\alpha}}{\mu_{\alpha}} \rho_{\alpha} x_{\alpha,i} (\nabla p_{\alpha} + \rho_{\alpha} g \nabla Z) \right\} + \sum_1 \left(\sum_{\alpha} \rho_{\alpha} x_{\alpha,i} q_{\alpha} \right)^1 = 0 \quad (1)$$

Where:

ϕ : porosity of the rock

S_{α} : saturation of phase α (oil, water, and LOHC)

ρ_{α} : density of phase α

$x_{\alpha,i}$: mole fraction of component i in phase α

k : absolute permeability tensor

$k_{r\alpha}$: relative permeability of phase α

μ_{α} : viscosity of phase α

p_{α} : pressure in phase α

g : gravitational acceleration

z : depth

This equation accounts for mass accumulation, Darcy flow, and source/sink contributions for each component in all active fluid phases. The LOHC (Toluene) is modeled as a separate component with appropriate phase behavior, allowing for its tracking during injection, soaking, and withdrawal periods. Capillary pressure and relative permeability between the water, oil, and LOHC phases are modeled using the (Brooks and Corey, 1964) correlations. The relative permeability for each phase is calculated by Eqs. (2–3).

$$k_{rw} = k_{rw0} \left(\frac{S_w - S_{wi}}{1 - S_{wi} - S_{or}} \right)^{n_w} \quad (2)$$

$$k_{ro} = k_{ro0} \left(\frac{1 - S_w - S_{or}}{1 - S_{wi} - S_{or}} \right)^{n_o} \quad (3)$$

Where k_{rw} refers to the relative permeability of the brine, k_{ro} denotes the relative permeability of the fluid, S_{wi} refers to irreducible water saturation, S_{or} indicates the residual oil saturation, k_{rw0} and k_{ro0} represent end point relative permeabilities, n_w and n_o represent Corey exponents for oil and water. Brooks and Corey (1964) model is also used to calculate the capillary pressure (P_c). The equation to get P_c is given by Eq. (4),

$$P_c = P_{ec} \times \left(\frac{S_w - S_{wi}}{1 - S_{wi} - S_{or}} \right)^{-\frac{1}{\lambda}} \quad (4)$$

Where λ represents the index for pore size distribution and P_{ec} is the capillary entry pressure. Figure 2 shows the relative permeability and capillary pressure curves.

Results and Discussions

The compositional reservoir simulation results provide crucial insights into the performance of Toluene as a LOHC under cyclic injection–withdrawal operations in a heterogeneous depleted sandstone reservoir. The minimum bottomhole pressure (BHP) constraint of 2500 psi was imposed to ensure safe operational limits during production and maximum BHP of 3250psi was employed during injection to ensure reservoir integrity. Figure 1 illustrates the operational profile of the cyclic LOHC injection–production system over 15 annual cycles. Each cycle begins with a 5-month injection phase, during which Toluene is injected at a constant rate of 2000 bbl/day, followed by a 2-month soaking period (flatline at zero), and then a 5-month production phase, during which LOHC and oil are withdrawn. During the production phase, the LOHC withdrawal rate initially spikes and gradually declines due to pressure drawdown and reservoir dynamics, while the total fluid production rate remains close to the injection rate, indicating good volumetric balance. The shaded area under the red curve highlights the LOHC contribution within the produced fluids. The consistency in cyclic patterns and the repeatability of both injection and production trends across all 15 years confirm the operational stability and reversibility of the LOHC system. The narrow gap between total production and LOHC withdrawal also implies significant oil production in early cycles, aligning with observed incremental oil recovery trends.

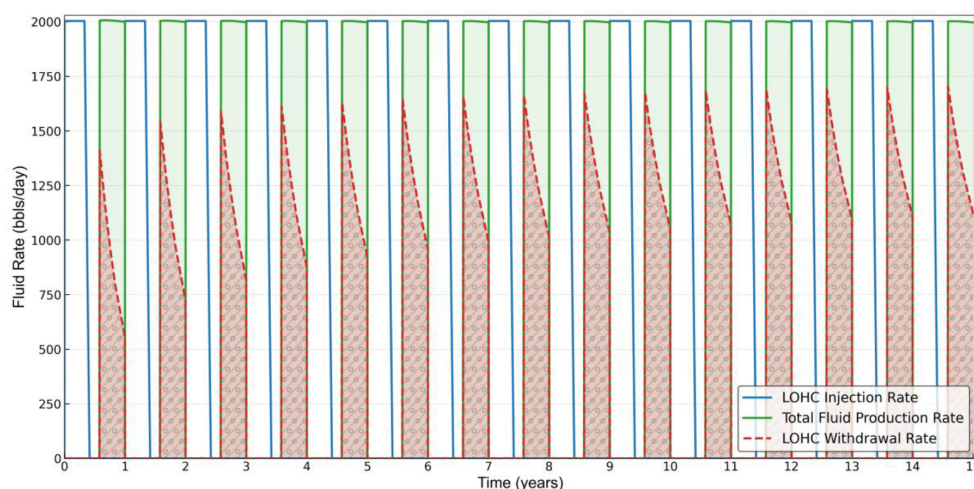


Figure 1—LOHC injection and withdrawal rates, along with the total fluid (oil + LOHC) production rates.

Figure 2 illustrates the cumulative incremental oil production and corresponding oil recovery factor over the course of 15 operational cycles. The figure clearly demonstrates a positive trend in oil recovery, where each subsequent cycle contributes additional incremental oil due to improved reservoir re-pressurization and favorable displacement dynamics. At the end of the 15th cycle, the cumulative incremental oil volume reaches a plateau, indicating that most of the recoverable residual oil has been mobilized. The overall incremental oil recovery factor approaches 35%, confirming the technical viability of using LOHC (Toluene) as a displacing agent in mature, depleted reservoirs. The sharp rise in recovery during the initial cycles can be attributed to the abundance of accessible residual oil and higher displacement efficiency. As cycles progress, the recovery rate gradually tapers off due to declining oil saturation and reduced mobility. This behavior aligns with typical EOR trends in heterogeneous formations, where early gains are more substantial than those in later stages.

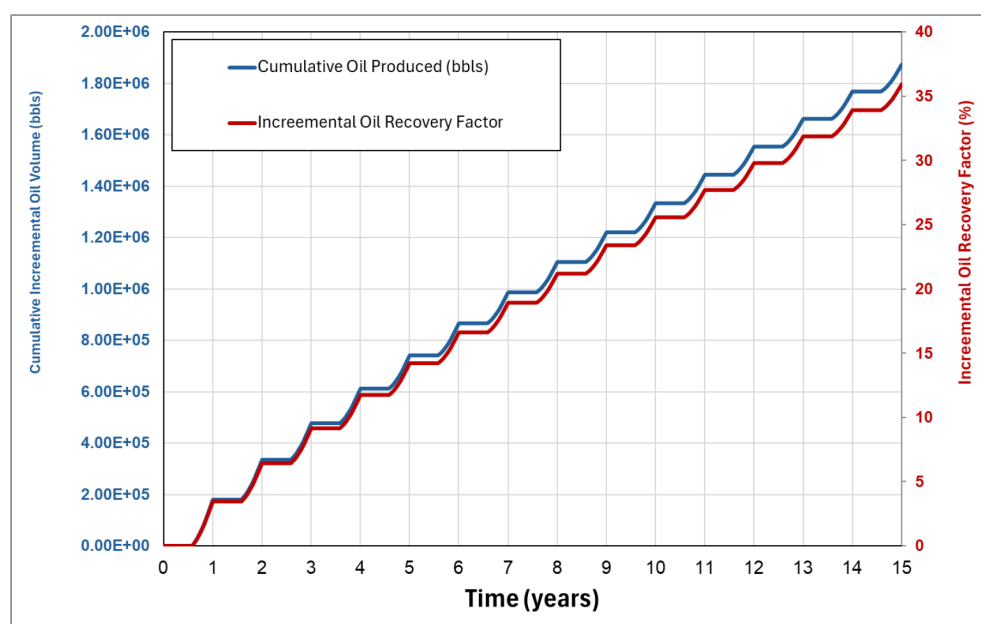


Figure 2—Cumulative Incremental Oil Volume with the incremental oil recovery factor.

Figure 3 presents the variation in LOHC (Toluene) recovery factor and oil cut over 15 injection–production cycles. The LOHC recovery factor shows a steady increase across the cycles, reaching

approximately 75% by the end of the 15th cycle. This indicates that a significant portion of the injected Toluene was successfully recovered during each withdrawal phase, affirming the reversibility and reusability of LOHC for subsurface hydrogen storage. The oil cut, which represents the fraction of oil in the total produced fluids, is observed to be high in the initial cycles and gradually declines in the later stages. This behavior reflects typical reservoir depletion patterns: early cycles mobilize a greater amount of residual oil due to favorable saturation and pressure gradients, while subsequent cycles yield diminishing returns as oil becomes less accessible. The combination of high LOHC recovery and decreasing oil cut demonstrates the dual functionality of the process—efficient hydrogen carrier storage with incremental oil recovery. Furthermore, the results validate the operational stability of repeated LOHC cycling, showing no signs of injectivity loss or breakthrough of unwanted phases such as water.

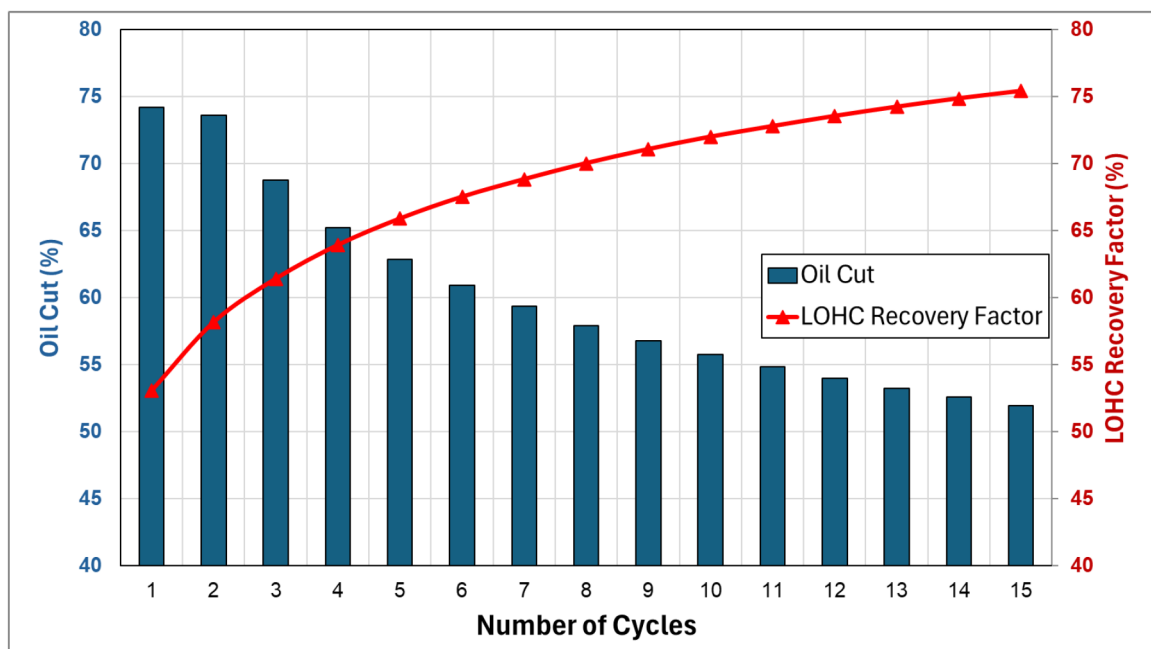


Figure 3—LOHC recovery factor and oil cut with the number of cycles

Figure 4 displays the well bottomhole pressure (BHP) trends for the injection well, production well, and the average reservoir pressure over the 15 cyclic operational periods. Each cycle comprises three phases: Toluene injection, soaking, and withdrawal. The figure illustrates the characteristic pressure fluctuations associated with each operational phase. During the injection phase, the BHP of the injection well increases as Toluene is introduced at a constant rate, maintaining pressures safely above the minimum constraint of 2000 psi and maximum constraint of 30% of initial reservoir pressure, thereby ensuring mechanical integrity of the reservoir and caprock. The soaking period shows a brief plateau or slight pressure redistribution due to fluid equilibration and re-pressurization within the reservoir. In contrast, the production well BHP shows a declining trend during each withdrawal period, corresponding to fluid extraction. These pressure drawdowns remain within the safe operational range, indicating no excessive reservoir depletion or risk of gas breakthrough or water coning. The stability and repeatability of the BHP profiles across all cycles confirm the robust injectivity and productivity performance of the reservoir. This behavior demonstrates that the formation can accommodate multiple LOHC injection–production cycles without significant pressure degradation, making it suitable for long-term reversible hydrogen storage and enhanced oil recovery. The average reservoir pressure (black line) demonstrates a gradual decline over time due to cumulative fluid withdrawal. However, the decline is controlled and linear, indicating that the reservoir maintains pressure support throughout all 15 cycles. The consistent separation between injection and production BHPs, coupled

with the stability of the reservoir pressure trend, highlights the efficiency and robustness of the cyclic LOHC operation. The absence of sharp pressure crashes or buildup over time confirms that the reservoir is well-suited for long-term cyclic hydrogen storage and enhanced oil recovery without encountering pressure-related integrity issues.

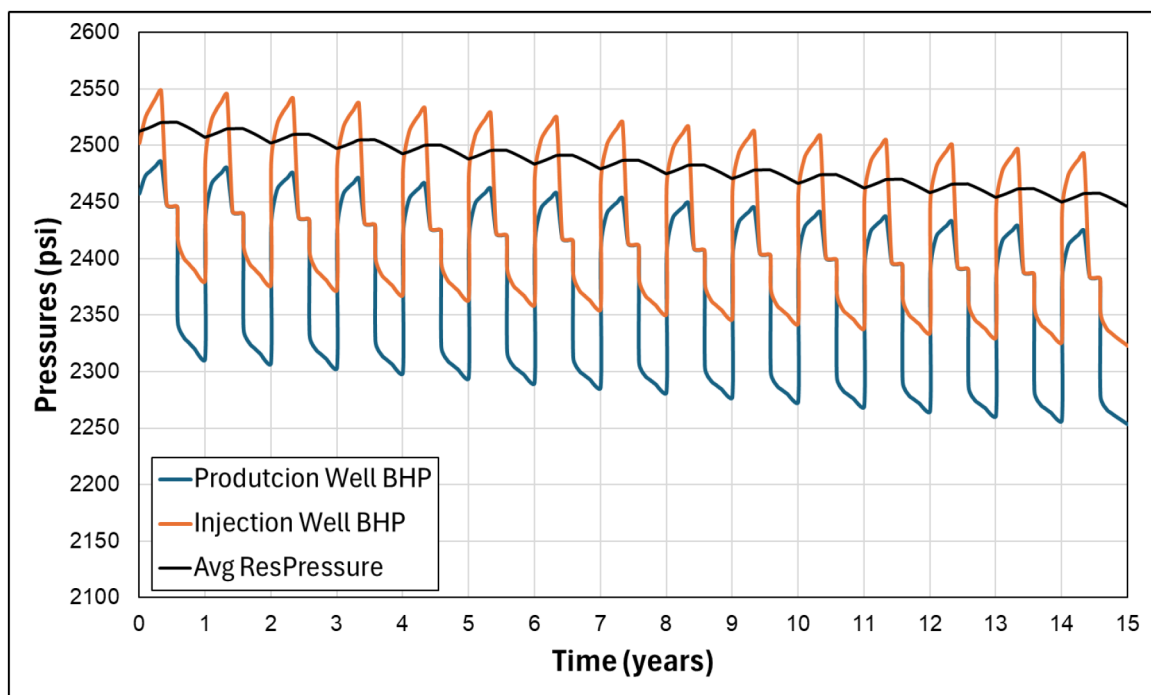


Figure 4—Bottomhole pressure (BHP) profiles of the injection and production wells, along with the average reservoir pressure, over 15 LOHC injection–production cycles.

Conclusions

This study demonstrates the technical feasibility of using Liquid Organic Hydrogen Carriers, specifically Toluene, for dual-purpose underground storage and enhanced oil recovery in a depleted sandstone oil reservoir. The simulation results confirm that the selected field-scale case offers a viable pathway to integrate hydrogen storage infrastructure with reservoir reactivation. Over 15 simulated cyclic injection–production years, the reservoir successfully maintained pressure levels, ensuring operational safety and mechanical integrity. The optimized LOHC injection strategy achieved an average recovery factor of approximately 75%, validating the effectiveness of cyclic storage and withdrawal. Additionally, the LOHC injection process facilitated significant residual oil mobilization, resulting in an incremental oil recovery factor of ~35%. The simulation also indicated extremely low water production, with water cut consistently remaining below 0.001%, underscoring the selectivity and efficiency of the displacement mechanism. These findings highlight a novel opportunity to couple hydrogen economy objectives with enhanced hydrocarbon production, offering both environmental and economic advantages.

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List of Symbols

BHP Well Bottomhole pressure
CMG Computer Modeling Group

EOR Enhanced Oil Recovery
 LOHC Liquid Organic Hydrogen Carrier
 MCH Methylcyclohexane
 Tol Toluene
 UHS Underground Hydrogen Storage

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